

Question	Answer	Marks
5(a)(i)	straight line through the origin shows that a is proportional to x	B1
	negative gradient shows that a is always in the opposite direction to x	B1
5(a)(ii)	$a_0 = \omega^2 x_0$	C1
	$\omega = 2\pi / T$	C1
	$T = 2\pi \sqrt{(x_0 / a_0)}$ $= 2\pi \sqrt{(1.2 / 13)}$ $= 1.9 \text{ s}$	A1
5(b)(i)	loss of energy of oscillations	B1
	due to <u>resistive</u> force(s)	B1
5(b)(ii)	line starting from $x = \pm 1.2 \text{ cm}$ at $t = 0$	B1
	line starting from non-zero value of x from $t = 0$ to $t = 2T$ that is entirely either above or below the t -axis	B1
	curve from $t = 0$ starting from non-zero x value, with both magnitude of x value and magnitude of gradient continuously decreasing	B1

9702/42

Cambridge International AS & A Level – Mark Scheme

October/November 2025

PUBLISHED

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